



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

FURTHER MATHEMATICS
PAPER 2

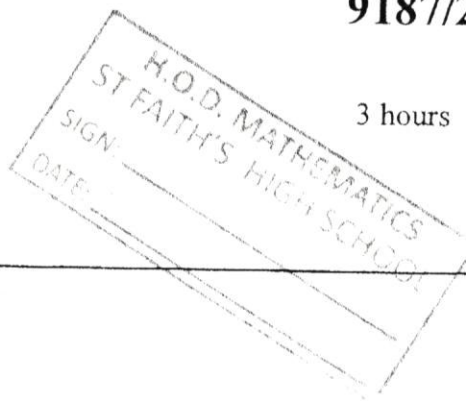
9187/2

NOVEMBER 2009 SESSION

3 hours

Additional materials:

Answer paper
Graph paper
List of Formulae



TIME 3 hours

INSTRUCTIONS TO CANDIDATES

Write your name, Centre number and candidate number in the spaces provided on the answer paper/answer booklet.

There is no restriction on the number of questions which you may attempt.

If a numerical answer cannot be given exactly, and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given correct to 2 significant figures.

If a numerical value for g is necessary, take $g = 9.81 \text{ ms}^{-2}$.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

The total number of marks for this paper is 120.

Within each section of the paper, questions are printed in the order of their mark allocations and candidates are advised, within each section, to attempt questions sequentially.

The use of an electronic calculator is expected, where appropriate.

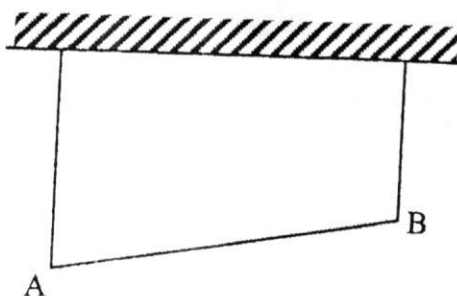
You are reminded of the need for clear presentation in your answers.

This question paper consists of 8 printed pages.

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Section (a) : Mechanics

1



The diagram shows a non-uniform rod AB of length l hanging in equilibrium and suspended by two vertical elastic strings. The strings have the same natural length and the same modulus of elasticity. In the equilibrium position shown, the extension of the string attached to A is twice that of the string attached to B.

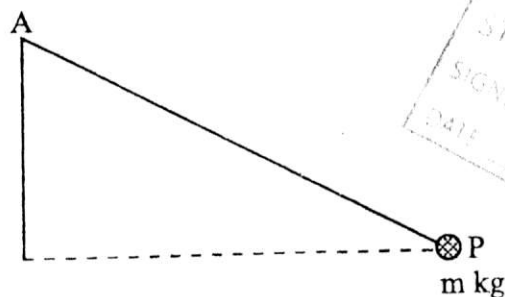
Show that the distance of the centre of gravity of the rod from A is $\frac{l}{3}$. [3]

2

A car of mass 1.2 tonnes is moving along a straight horizontal road under the action of a constant driving force of magnitude T newtons. The resistance to motion is $4v^2$ newtons, where v is the speed of the car.

- (a) Given that the acceleration is 4 ms^{-2} when $V = 16$, show that $T = 5824 \text{ N}$. [2]
- (b) Find the power of the car's engine when $v = 20$. [1]
- (c) Find the distance covered by the car as its speed increases from 10 ms^{-1} to 20 ms^{-1} . [4]

3



The diagram shows a conical pendulum, consisting of a light inextensible string which has one end attached to a fixed point A. A particle P of mass m is attached to the other end of the string. The particle P moves with constant speed completing 2 revolutions every second while the tension in the string is $2mg$.

Find, to the nearest centimetre

(a) the radius of the circular path of P, [5]

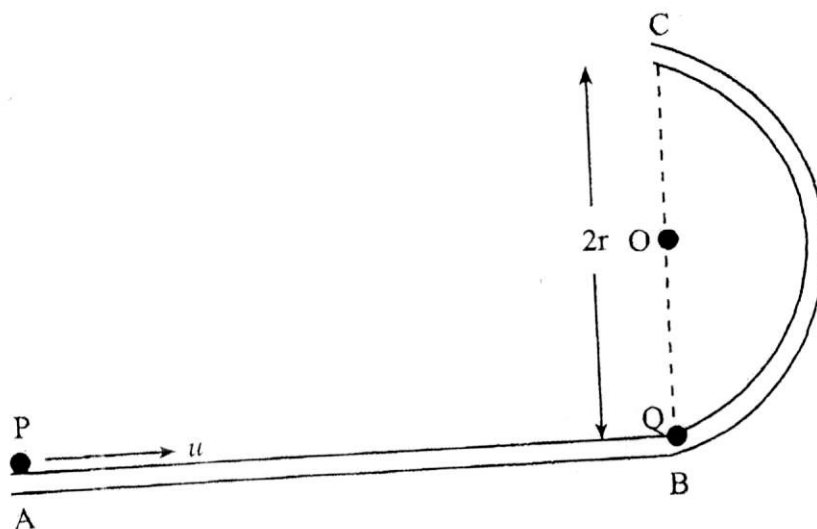
(b) the length of the string. [2]

4

A uniform rectangular sheet of metal ABCD of mass 5 kg is suspended from A. In equilibrium, AB, which has length 0.15 m, is inclined at 40° to the Vertical.

(a) Find the length of AD. [2]

(b) A metal ring of mass 2 kg is now suspended from D. Find the angle between AB and the vertical in the new equilibrium position. [6]

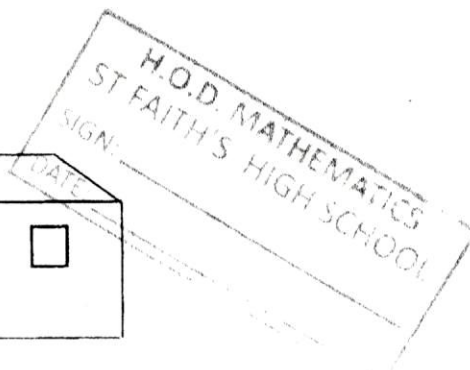
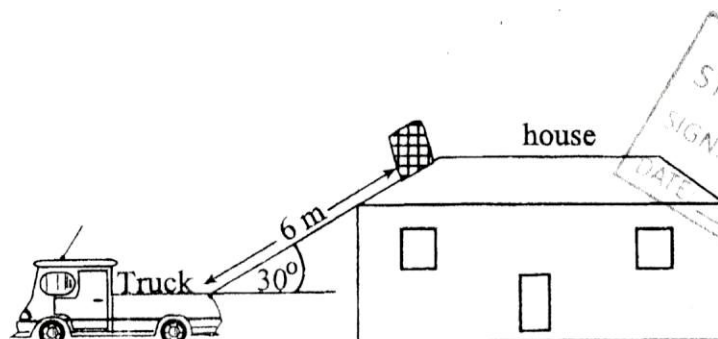


In a mechanics experiment small spheres P and Q, of masses $2m$ and m respectively, slide along a smooth groove in a rail. The rail, which is fixed in a vertical plane is in the form of a horizontal straight line AB joined to a semi circular arc BC of radius r and centre O, as shown in the diagram. Sphere P is projected with speed u along AB towards sphere Q which is initially at rest at the point B. There is a direct collision between P and Q, immediately after which the speed of Q is $\frac{5u}{4}$.

- (a) Find the speed of P immediately after the collision. [2]
- (b) Show that for Q to remain in contact with the groove on the complete arc BC, [7]

$$u \geq \sqrt{\left(\frac{16ga}{5}\right)}.$$

- 6 A carpenter removes identical roof tiles from the roof of a house by sliding them down a plank into the trailer of a truck. The plank is inclined at 30° to the horizontal [see diagram]. He releases a tile of mass 5 kg from rest. When the tile reaches the trailer of the truck its speed is 5 ms^{-1} , having slid 6 m down the plank.



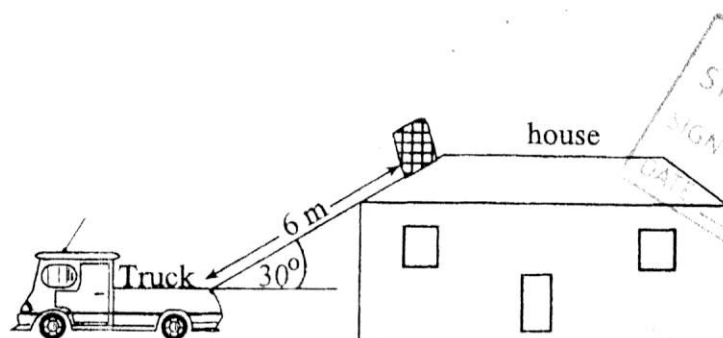
- (a) Calculate
- (i) the potential energy, lost by the tile, [1]
 - (ii) the frictional force which acts on the tile while it is sliding, [3]
 - (iii) the coefficient of friction between the tile and the plank. [2]
- (b) The carpenter pushes another tile down the plank, giving it an initial speed of 2 ms^{-1} .
- Find the speed of the tile when it has slid 6 m . [4]

- 7 A particle P of mass m is attached at one end of a light elastic string of natural length l and modulus of elasticity $12 mg$. The other end of the string is attached to a fixed point O.

- (a) Initially, P is held at O and then allowed to fall freely.
- (i) Find its speed when the string first becomes taut. [2]
 - (ii) Show that P first comes instantaneously to rest at a distance $\frac{3l}{2}$ below O. [5]
- (b) If, instead, P is allowed to hang freely in equilibrium at point E below O, determine the length of the string OE. [2]
- (c) The particle is now pulled down a further distance $\frac{l}{4}$ below E and released.
- The displacement measured downwards from E is x at time t after release.
- Show that the ensuing motion is simple harmonic with period

$$T = \pi \sqrt{\frac{l}{3g}}. \quad [4]$$

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The displacement measured downwards from E is x at time t after release.

Show that the ensuing motion is simple harmonic with period

$$T = \pi \sqrt{\frac{l}{3g}}. \quad [4]$$

- (d) Show that when the particle is a distance l below O, its speed, V is given by $V^2 = \frac{2}{3}gl$. [3]

Section (b): Statistics

- 8 The pressure P , and volume V of a fixed mass of gas are believed to be related by an equation of the form $pv^c = d$, where c and d are constants. [1]
- (a) Show that $pv^c = d$ may be reduced to a linear model of $\ln v$ against $\ln p$.
- (b) By putting $x = \ln p$ and $y = \ln v$, the results based on six experiments are summarised as:

$$\begin{aligned} \sum x &= 2.420, & \sum y &= -1.708, & \sum x^2 &= 3.171 \\ \sum y^2 &= 1.561 \text{ and } & \sum xy &= -2.224. \end{aligned}$$

Find the value of c and of d correct to three significant figures. [5]

- 9 The mass of a breakfast cereal is known to be normally distributed with mean μ grammes and standard deviation 0.92 grammes. It is required to have a 98% symmetric confidence interval with a total width of less than 0.53 grammes. [6]
- Find the least number of packets in a sample needed to achieve this.

- 10 A discrete random variable X has the following probability distribution;

X	0	1	2	3
$P(X = x)$	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	$\frac{1}{8}$

Find the mean and the variance. [7]

- 11 A bakery owner claims that the average mass of a loaf of bread produced by his company is 700 grammes. This is however disputed by the Standards Association of Zimbabwe as an over-estimate. A random sample of 10 loaves is taken and the mass of each loaf measured. The results are shown below.

Mass (grammes)	650	680	690	710	720
Frequency	2	3	2	2	1

- (i) Carry out a suitable test to test whether there are reasonable grounds for the Association to dispute the bakery owner's claim. [8]

- 12 A box contains n oranges. The probability that an orange is underweight is $\frac{1}{3}$. A box is said to be sub-standard if it contains 40 or more underweight oranges. Use a suitable approximation to estimate the greatest value of n given that the box is sub-standard is to be less than 0.01. [10]
- 13 Incoming calls arrive at a switchboard at random times, at an average rate of two per hour.
- (a) Find the probability that there are,
- (i) at least two calls in a period of three hours, [3]
 - (ii) more than three calls in a period of 30 minutes, [3]
 - (iii) no calls are received in a period of 10 minutes. [2]
- (b) Outgoing calls are received independently of incoming calls, at an average rate of three per hour.
- Find the probability that a combined number of at most two calls are received in a period of 30 minutes. [3]
- 14 In a seed germination test, 600 seeds were planted in 100 rows of 6. The number of seeds that germinated in each row was counted and the results are shown in the table below.

Number of seeds germinating per row	0	1	2	3	4	5	6
Number of rows	1	4	7	29	33	18	8

- (a) Calculate the mean number of seeds germinating per row. [1]
- (b) Find the frequencies which would be associated with a Binomial distribution having mean found in (a). [4]
- (c) Carry out a χ^2 goodness of fit test to test the validity of Binomial model at 5% significance level. [7]